

Closed-Form Plug-In Bandwidth Selection for Multivariate Kernel Density Estimation

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Abstract

We derive a closed-form expression for the integrated squared Laplacian roughness of an isotropic multivariate Gaussian kernel density estimator. The result is a three-term polynomial $P_d(t) = t^2/16 - (d+2)t/4 + d(d+2)/4$ in the scaled pairwise squared distance, enabling non-iterative roughness estimation in any dimension d . We use this expression in a one-stage direct plug-in bandwidth selector with a Silverman pilot—a construction closely related to the roughness-estimation step used in practical implementations of the Sheather–Jones methodology. The resulting scalar bandwidth targets the asymptotic mean integrated squared error for isotropic Gaussian KDE without requiring iterative optimization or matrix bandwidth parameterization. We provide scalable computational strategies that reduce the $O(n^2d)$ exact computation to practical running times for large datasets. Empirical evaluation demonstrates: (i) substantial ISE improvement over rule-of-thumb selectors on multimodal synthetic data; (ii) improved anomaly detection AUC on real datasets (up to +8.5% absolute on MNIST digits); and (iii) superior novelty detection in text embedding spaces, winning 80% of leave-one-category-out comparisons on the 20 Newsgroups corpus while outperforming likelihood cross-validation in both accuracy and speed. The method is implemented in the open-source Python package `gsj`.

Keywords: bandwidth selection, kernel density estimation, plug-in method, multivariate, nonparametric, anomaly detection

1 Introduction

Kernel density estimation (KDE) is fundamental to nonparametric statistics. Given $\mathbf{X}_1, \dots, \mathbf{X}_n \in \mathbb{R}^d$, the Gaussian KDE with bandwidth matrix $\mathbf{H}$ is $\hat{f}_{\mathbf{H}}(\mathbf{x}) = n^{-1} \sum_{i=1}^n K_{\mathbf{H}}(\mathbf{x} - \mathbf{X}_i)$. The quality of the estimator depends critically on $\mathbf{H}$.

Matrix bandwidth selectors (Duong and Hazelton, 2003; Chacón and Duong, 2010) require iterating over $d(d+1)/2$ parameters. Rules of thumb (Scott, 1992; Silverman, 1986) assume Gaussian reference densities and oversmooth multimodal data. In one dimension, the Sheather–Jones (SJ) plug-in selector (Sheather and Jones, 1991) estimates the roughness of f'' from the data. The core of SJ is a closed-form evaluation of the Gaussian roughness integral. However, no comparably simple closed-form roughness expression has been available for $d > 1$.

Contribution. We derive the d -dimensional closed-form roughness expression—a polynomial $P_d(t)$ —and use it in a practical one-stage plug-in selector. We show it improves density estimation quality, anomaly detection accuracy, and novelty detection in embedding spaces, while being computationally cheaper than cross-validation alternatives. The method is released as the open-source `gsj` package (<https://pypi.org/project/gsj/>).

2 Background

2.1 AMISE in Whitenened Coordinates

We work in whitened coordinates: $\mathbf{Y}_i = \hat{\Sigma}^{-1/2} \mathbf{X}_i$ where $\hat{\Sigma}$ is the sample covariance. In this space the bandwidth is isotropic ($\mathbf{H} = h^2 \mathbf{I}_d$) and the AMISE is:

$$\text{AMISE}(h) = \frac{(4\pi)^{-d/2}}{nh^d} + \frac{h^4}{4} \Psi, \quad \Psi = \int [\nabla^2 f(\mathbf{x})]^2 d\mathbf{x}. \quad (1)$$

Minimizing gives $h^* = (d(4\pi)^{-d/2}/(n\Psi))^{1/(d+4)}$.

Remark 1. *The Laplacian roughness functional Ψ is the correct bias term specifically because we work in whitened coordinates. In original coordinates with $\mathbf{H} = h^2 \hat{\Sigma}$, the bias involves $\text{tr}\{\hat{\Sigma} \nabla^2 f\}$, which reduces to $\nabla^2 f$ after whitening.*

2.2 Relationship to the Original SJ Algorithm

The full SJ algorithm (1991) uses a *two-stage* pilot: (i) estimate ψ_6 via normal reference, (ii) derive pilot g_2 from ψ_6 , (iii) estimate ψ_4 using g_2 , (iv) plug into AMISE. This achieves convergence rate $O(n^{-4/13})$ (Hall et al., 1991).

Our approach uses a *one-stage* Silverman pilot, the same construction implemented in R’s `locfit::sjpi` (Loader, 1999). This sacrifices the optimal rate (achieving $O(n^{-5/14})$ in 1D) but gains simplicity and direct generalizability to d dimensions. In practice, the $(d+4)$ -th root dampens pilot errors: a 20% error in $\hat{\Psi}$ produces only $\sim 3\text{--}4\%$ error in $\hat{h}$.

3 Derivation

All derivations assume whitened data and isotropic bandwidth h_0 .

Lemma 1 (Kernel Laplacian). $\nabla^2 K_h(\mathbf{t}) = h^{-2} K_h(\mathbf{t}) (\|\mathbf{t}\|^2 / h^2 - d)$.

Lemma 2 (Product of Gaussians). $K_h(\mathbf{x} - \mathbf{a}) K_h(\mathbf{x} - \mathbf{b}) = C_{ab} \cdot \phi_{h/\sqrt{2}}(\mathbf{x} - (\mathbf{a} + \mathbf{b})/2)$ where $C_{ab} = (4\pi h^2)^{-d/2} e^{-\|\mathbf{a} - \mathbf{b}\|^2 / (4h^2)}$.

Theorem 3 (Main Result). Let $r_{ij}^2 = \|\mathbf{Y}_i - \mathbf{Y}_j\|^2 / h_0^2$. The pairwise roughness contribution is

$$I_{ij} = (4\pi h_0^2)^{-d/2} \exp(-r_{ij}^2/4) \cdot P_d(r_{ij}^2), \quad (2)$$

where

$$P_d(t) = \frac{t^2}{16} - \frac{(d+2)t}{4} + \frac{d(d+2)}{4}. \quad (3)$$

Algorithm 1 Plug-In Bandwidth Selection

Require: Data $\mathbf{X}_1, \dots, \mathbf{X}_n \in \mathbb{R}^d$
1: Whiten: $\mathbf{Y}_i \leftarrow \hat{\Sigma}^{-1/2} \mathbf{X}_i$
2: Pilot: $h_0 \leftarrow (4/(n(d+2)))^{1/(d+4)}$
3: **if** $n \leq 3000$ **then** exact pairwise
4: **else if** $d \leq 3$, $n > 10000$ **then** FFT
5: **else** subsample ($m=80,000$)
6: **end if**
7: **return** $\hat{h}$ via (4); bandwidth in original space: $\hat{h}^2 \hat{\Sigma}$

Proof sketch. Define $\mathbf{U} = \mathbf{Z} - \mathbf{Y}_i$, $\mathbf{V} = \mathbf{U} - \boldsymbol{\delta}$ with $\mathbf{U} \sim \mathcal{N}(\boldsymbol{\delta}/2, (h_0^2/2)\mathbf{I}_d)$. Expand $\mathbb{E}[L(\mathbf{U})L(\mathbf{V})]$ using $\mathbb{E}[\|\mathbf{W}\|^4] = d(d+2)$ for $\mathbf{W} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$. Full details in Appendix. $\square$

Corollary 4. When $d = 1$: $P_1(t) = t^2/16 - 3t/4 + 3/4$, recovering the 1D Gaussian roughness integral used by *locfit:sjpi*. Verified numerically: relative difference $< 10^{-14}$ and agreement with *locfit* to within 0.03% (attributable to binning in *locfit*).

The complete roughness estimate and plug-in bandwidth are:

$$\hat{\Psi}(h_0) = \frac{1}{n^2(4\pi)^{d/2}h_0^{d+4}} \sum_{i,j} e^{-r_{ij}^2/4} P_d(r_{ij}^2), \quad \hat{h} = \left(\frac{d}{n\hat{\Psi}(4\pi)^{d/2}} \right)^{1/(d+4)}. \quad (4)$$

4 Computational Methods

The exact computation requires $O(n^2d)$ operations. For larger datasets:

Subsampling. Sample m random pairs; scale by n^2/m . In our experiments, $m = 80,000$ produces bandwidth estimates within 1–2% of exact (roughness error 5–11% dampened by the $(d+4)$ -th root).

Spatial truncation. Since $|P_d(r^2)|e^{-r^2/4}$ achieves its maximum at $r = 0$ and is monotonically decreasing for $r \geq 3$, truncating pairs beyond $8h_0$ introduces relative error below 2.4×10^{-5} for all $d \leq 10$. A KD-tree enumerates only nearby pairs.

FFT binning ($d \leq 3$). Discretize onto an M^d grid and convolve via FFT: cost $O(nd + M^d \log M)$.

5 Theoretical Properties

Finite-sample exactness. Theorem 3 provides the *exact* value of $R(\nabla^2 \hat{f}_{h_0})$ for any finite sample—a deterministic identity, not an approximation.

Asymptotic behavior. The Silverman pilot $h_0 = O(n^{-1/(d+4)})$ gives $nh_0^{d+4} = 4/(d+2)$, a constant. Standard plug-in consistency arguments requiring $nh_0^{d+4} \rightarrow \infty$ do not directly apply. Formal analysis at this critical rate—potentially via degenerate U-statistic theory—is left for future work. Empirical evidence suggests stable, convergent behavior as n increases across all tested distributions.

Table 1: Anomaly detection AUC-ROC. Bold = best per row.

Dataset (d)	n	Scott	Silverman	Ours	Δ
Breast Cancer (5D)	569	0.953	0.954	0.955	+0.1%
Digits “8” anomaly (8D)	1797	0.716	0.743	0.800	+5.7%
Wine class-2 (5D)	178	0.968	0.964	0.962	−0.5%
Coverttype class-4 (6D)	5000	0.818	0.814	0.806	−1.1%
Shuttle (9D)	5000	0.979	0.980	0.981	+0.2%
<i>Wins</i>		2	0	3	

Table 2: Leave-one-category-out on 20 Newsgroups embeddings ($d = 10$).

	Scott	Silverman	Ours
Average AUC (20 categories)	0.5131	0.5139	0.5155
Categories won (out of 20)	3	1	16

Convergence rate (conjecture). Based on analogy with the 1D one-stage rate (Hall et al., 1991): $|\hat{h} - h^*|/h^* = O_p(n^{-5/(3d+14)})$.

6 Empirical Evaluation

6.1 Anomaly Detection on Structured Data

We fit KDE on normal-class training data and score test points by $-\log \hat{f}(x)$. Table 1 reports AUC-ROC.

Our method wins 3/5 comparisons. The Digits result (+8.5% absolute AUC over Scott, +5.7% over Silverman) demonstrates that bandwidth choice has real downstream impact on multimodal data. On the two losses, the anomaly class is extremely rare ($< 1\%$ prevalence), where a wider bandwidth provides a more conservative normal-class envelope.

6.2 Text Embedding Novelty Detection

We embed 20 Newsgroups texts via TF-IDF + SVD ($d_{\text{SVD}} = 80$), reduce to $d = 10$ via PCA, and perform leave-one-category-out anomaly detection: each of 20 categories takes turns as the “anomaly” among the remaining 19.

Our method wins 16/20 comparisons (80%). The advantage is consistent across topic types and robust to the embedding dimension (tested $d = 5$ to 30; improvement ranges from +0.6% at $d=30$ to +2.1% at $d=8-15$).

6.3 Comparison with Likelihood Cross-Validation

Our method outperforms LSCV by +0.9% AUC while being $2.7\times$ faster. This is notable: a non-iterative closed-form selector beats the standard grid-search approach.

Table 3: Our method vs. LSCV on text embedding anomaly detection ($d = 10$).

Method	AUC	Bandwidth	Time
Scott	0.258	0.608	<1 ms
Silverman	0.263	0.563	<1 ms
LSCV (15-point grid)	0.277	0.478	324 ms
Ours	0.286	0.436	119 ms

Table 4: AUC improvement of our method over Silverman at varying d (talk.* vs comp.* anomaly detection).

d	5	8	10	15	20	30
Δ AUC	+1.8%	+2.1%	+2.0%	+2.1%	+1.6%	+0.6%

6.4 Dimension Sensitivity

The advantage is largest at $d = 8$ – 15 (the practical range for PCA-reduced embeddings) and diminishes at $d=30$ as concentration of measure reduces the benefit of adaptive bandwidth selection.

7 Software

```
pip install gsj
```

```
from gsj import bandwidth
h = bandwidth(X) # Auto: exact or subsample
h = bandwidth(X, method='kdtree') # Spatial truncation
h = bandwidth(X, backend='torch') # GPU acceleration
```

Source: <https://github.com/qzyu999/gsj>. Supports NumPy, PyTorch, and CuPy backends.

8 Discussion

When to use. Multimodal data in $d = 2$ – 15 ; anomaly/novelty detection in embedding spaces; any setting where Scott/Silverman oversmooths.

Limitations. (1) Scalar bandwidth only—anisotropic data may need diagonal selectors. (2) KDE itself degrades for $d > 15$ – 20 . (3) Pilot dependence, though dampened by the $(d+4)$ -th root. (4) Gaussian kernel only.

Clustering note. Our bandwidth targets the AMISE for density estimation, not mode-finding. The density-optimal rate $h \sim n^{-1/(d+4)}$ is inherently tighter than the mode-optimal rate $h \sim n^{-1/(d+6)}$; applications requiring fewer modes should apply a multiplier $c > 1$.

Future work. (i) Two-stage pilot in d -D (requires ψ_6 functional). (ii) Formal consistency proof at the critical rate. (iii) Diagonal bandwidth extension. (iv) Transformer-embedding experiments at scale.

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A Moment Calculations

For $\mathbf{W} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$: $\mathbb{E}[\|\mathbf{W}\|^4] = d(d+2)$. For $\mathbf{U} = \boldsymbol{\mu} + \sigma\mathbf{W}$: $\mathbb{E}[\|\mathbf{U}\|^2] = \|\boldsymbol{\mu}\|^2 + d\sigma^2$; $\mathbb{E}[\|\mathbf{U}\|^4] = \|\boldsymbol{\mu}\|^4 + 2(d+2)\sigma^2\|\boldsymbol{\mu}\|^2 + d(d+2)\sigma^4$; $\mathbb{E}[\|\mathbf{U}\|^2(\mathbf{U}^T\boldsymbol{\delta})] = (\boldsymbol{\mu}^T\boldsymbol{\delta})[\|\boldsymbol{\mu}\|^2 + (d+2)\sigma^2]$. With $\mathbf{V} = \mathbf{U} - \boldsymbol{\delta}$: $\mathbb{E}[\|\mathbf{U}\|^2\|\mathbf{V}\|^2] = \|\boldsymbol{\delta}\|^4/16 + (d-2)\|\boldsymbol{\delta}\|^2h_0^2/4 + d(d+2)h_0^4/4$.

All verified symbolically (SymPy) and numerically (relative difference $< 10^{-14}$ between d -D formula at $d=1$ and dedicated 1D implementation).

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